INCH-POUND
MIL-DTL-15291/11C
21 January 2022
SUPERSEDING
MIL-DTL-15291/11B
6 February 2006

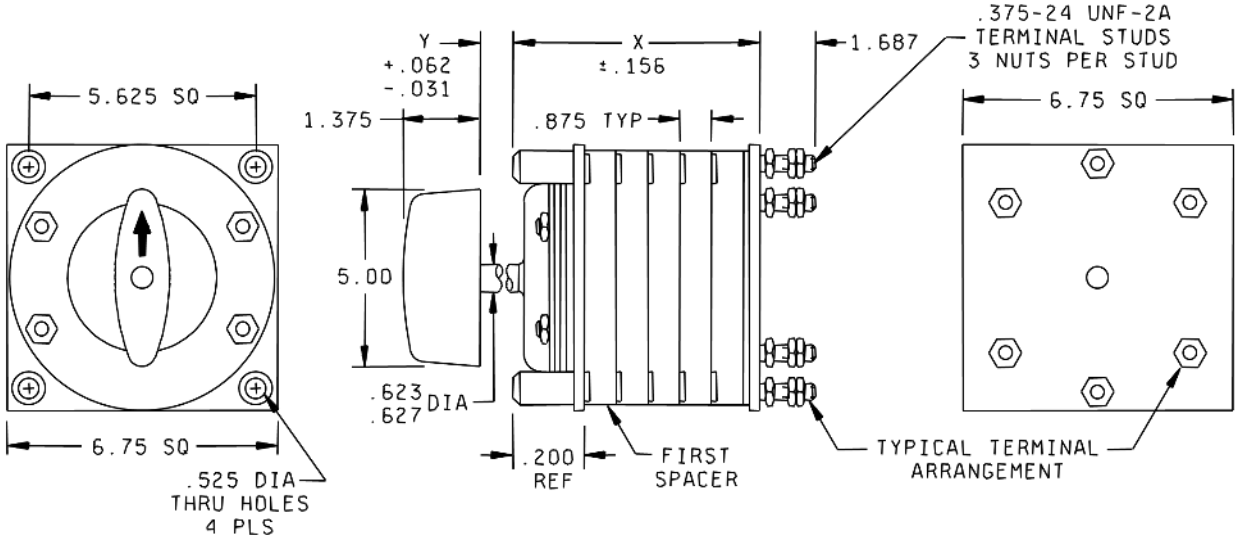
# DETAIL SPECIFICATION SHEET

## SWITCHES, ROTARY, SNAP ACTION CLASS 20SR FRONT MOUNTED, BACK CONNECTED

Inactive for new design after 17 April 1981. For new design, use MIL-DTL-15291/12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-DTL-15291.




NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .015.

FIGURE 1. Class 20SR switch (up to 6 studs).

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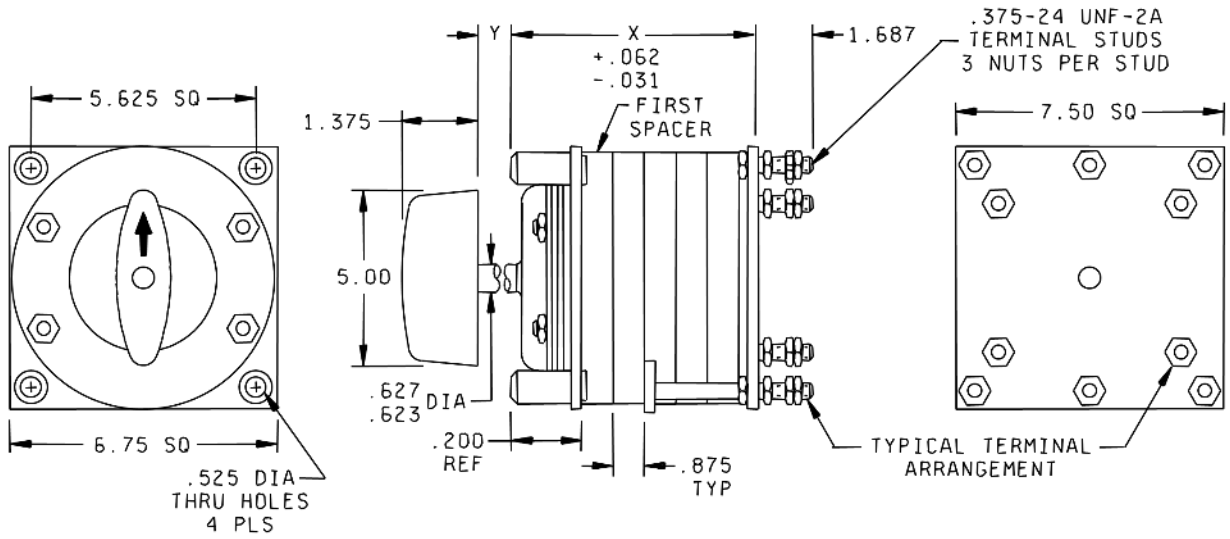



FIGURE 2. Class 20SR switch (up to 10 studs).

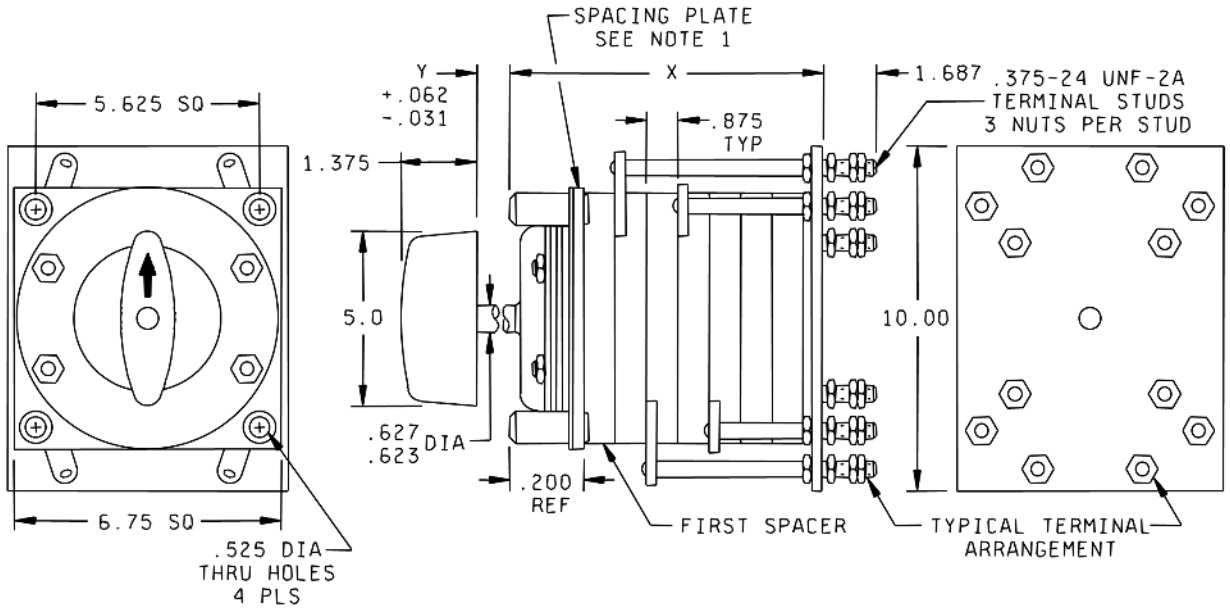
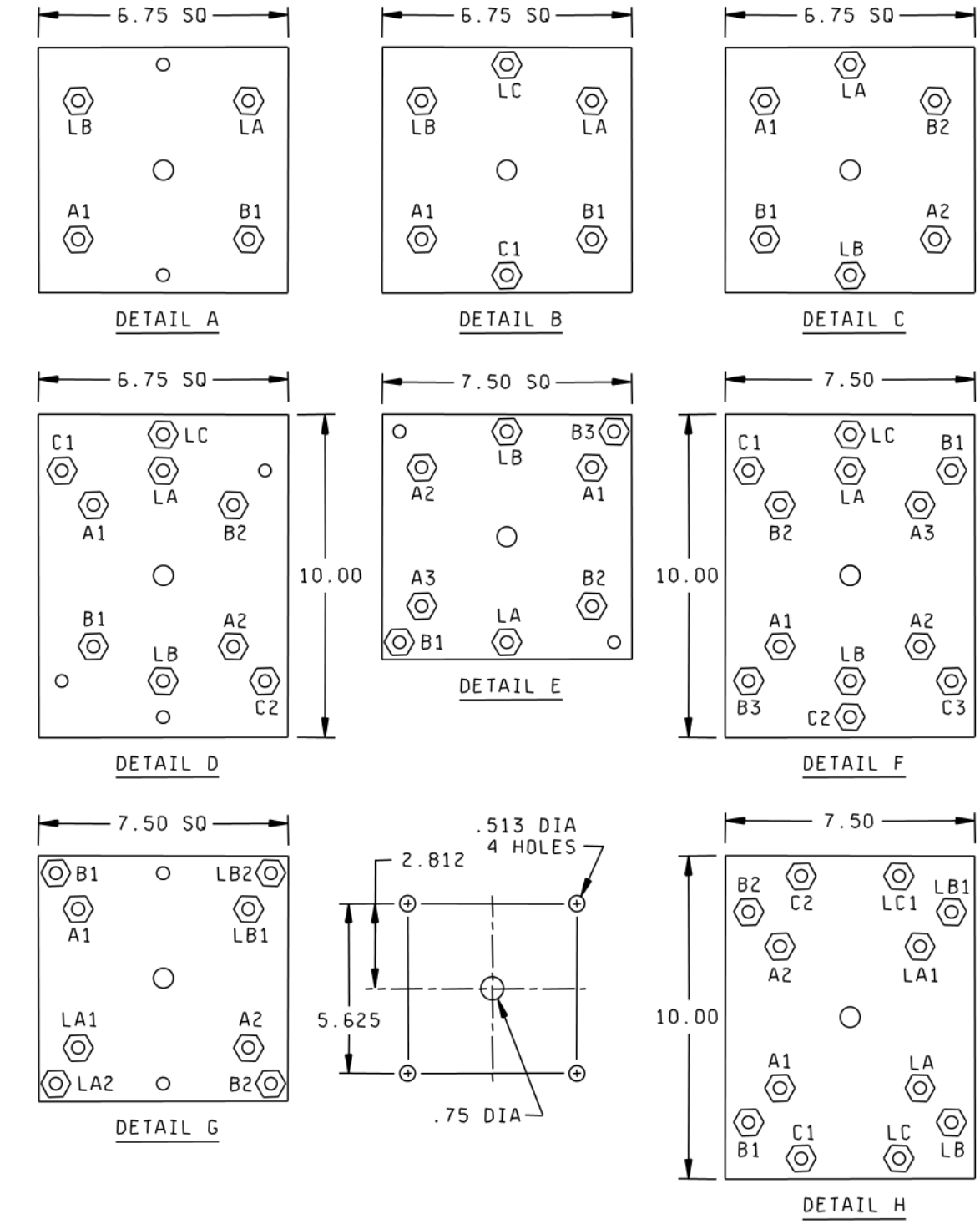



FIGURE 3. Class 20SR switch (up to 12 studs).



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NOTE: Terminal studs viewed from rear.

FIGURE 4. Mounting dimensions and terminal stud locations.



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TABLE I. Type and switching characteristics.

| M15291/11 DASH NO. TYPE DESIGNATION | DETAIL REF / TORQUE IN-LBS | DIM "X" / DIM "Y" | HANDLE OR SHAFT POSITION | CIRCUIT AND SPACER CONFIGURATION ROTOR POSITION, SPACER LOCATIONS, TERMINAL MARKING AND LOCATIONS<br>NO 1 | NO 2 | NO 3 | NO 4 | NO 5 | NO 6 | NO 7 | NO 8 | NO 9 | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| -001<br>20SR2A1 | FIG 1&4 DETAIL A<br>72 | 4.937<br>.437 | OFF<br>ON<br>ON<br>OFF | | LB<br>B1 | LA<br>A1 | | | | | | | |
| -002<br>20SR3A1 | FIG 1&4 DETAIL A<br>72 | 5.812<br>.437 | OFF<br>ON<br>ON<br>OFF | | LC<br>C1 | LB<br>B1 | LA<br>A1 | | | | | | |
| -003<br>20SR2B1 | FIG 1&4 DETAIL B<br>72 | 6.687<br>.437 | OFF<br>ON 2<br>ON 1<br>OFF | | LB | B2<br>B1 | LA | A1<br>A2 | | | | | |
| -004<br>20SR3B1 | FIG 3&4 DETAIL C<br>72 | 8.437<br>.437 | OFF<br>ON 2<br>ON 1<br>OFF | | LC<br>C2 | C1 | LB | B2<br>B1 | LA | A1<br>A2 | | | |
| -005<br>20SR2E1 | FIG 2&4 DETAIL D<br>72 | 6.687<br>.437 | OFF<br>ON 3<br>ON 1<br>ON 2 | | LB<br>B2 | B3<br>B1 | LA | A1<br>A2<br>A3 | | | | | |
| -006<br>20SR3E1 | FIG 3&4 DETAIL E<br>72 | 8.437<br>.437 | OFF<br>ON 3<br>ON 1<br>ON 2 | | LC<br>C3 | C1<br>C2 | LB | B1<br>B2<br>B3 | LA | A3<br>A2<br>A1 | | | |
| -007<br>20SR4F1 | FIG 2&4 DETAIL E<br>72 | 6.687<br>.437 | OFF<br>ON 2<br>ON 1<br>ON 1&2 | | LB2<br>B2 | LB1<br>B1 | A2<br>LA2 | A1<br>LA1 | | | | | |
| -008<br>20SR6F1 | FIG 3&4 DETAIL F<br>72 | 8.437<br>.437 | OFF<br>ON 2<br>ON 1<br>ON 1&2 | | LC1<br>C2 | LC<br>C1 | LB1<br>B2 | LB<br>B1 | LA1<br>A2 | LA<br>A1 | | | |



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REQUIREMENTS:

Applicable specification: MIL-DTL-15291.

Dimensions and mounting: See figures 1, 2, 3, and 4.

Switching characteristics: See table I.

Angle of throw: 90 degrees.

Switching action: Snap action, reciprocating.

Electrical and endurance ratings: See table II.

Stop strength (applicable to switches with stops): 100 inch-pounds.

Vibration: 50 Hz, MIL-STD-167-1.

Shock: High impact, MIL-DTL-901.

Contact resistance: 1.5 milliohms maximum.

Dielectric withstanding voltage: 2,000 V rms.

Insulation resistance: 200 megohms minimum.

Temperature rise: 50°C maximum.

Mounting bolts: (4), 1/2 diameter, length, and headstyle to suit application. Bolts not furnished.

Handle: M15291/12-100 per MIL-DTL-15291/12.

Terminal marking: Terminal marking shall be in accordance with details A through H, as specified in table I.



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TABLE II. Electrical and endurance ratings.

| Tests | Current (amperes) | Voltage (volts) | Electrical operations (number of operations) | Test rate (operations per minute) | De-energized operations (number of operations) |
|---|---|---|---|---|---|
| Alternating current (rms) | 200 | 500 | 6,000 | 8 | 4,000 |
| Direct current | 200 | 250 | 4,000 | 3 | --- |

GENERAL INFORMATION:

Switches not covered by specification sheets: Switches which are fabricated from standard parts, as used in qualified switches, but which do not comply with switches detailed herein with respect to circuit characteristics, switching action, mounting arrangements, and handle details may be acquired under this specification from contractors having qualification approval under this specification.

Extended ratings: Switches detailed herein have been tested and found satisfactory at the extended ratings listed in table III. Reduced life expectancy must be anticipated for switches used in these increased voltage or current levels. Tests under the conditions of table III are not required for qualification acceptance and they are not repeated routinely as for maintenance of qualification. Supplemental evaluations and tests applicable to particular circuit requirements are recommended.



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TABLE III. Extended ratings.

| Switching characteristics | AC 60 or 400 Hz | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 125 volts | | | | 250 volts | | | | 500 volts | | | |
| | Resistive or lamp load | | Inductive load 0.75 p.f. | | Resistive or lamp load | | Inductive load 0.75 p.f. | | Resistive or lamp load | | Inductive load 0.75 p.f. | |
| | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations |
| All | 200 | 6,000 | 200 | 6,000 | 200 | 6,000 | 200 | 6,000 | 200 | 6,000 | 200 | 6,000 |
| | DC | | | | | | | | | | | |
| | 120 volts | | | | 250 volts | | | | 350 volts | | | |
| | Resistive or Lamp load | | Inductive load[1] | | Resistive or Lamp load | | Inductive load[1] | | Resistive or Lamp load | | Inductive load[1] | |
| | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations |
| All | 200 | 4,000 | -- | -- | 200 | 4,000 | 200 | 4,000 | -- | -- | -- | -- |

1/ 0.025 henry for inductive circuit.





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APPLICATION AND ACQUISITION GUIDE: PIN and type designation cross reference shall be as shown in table IV.

TABLE IV. Application and acquisition guide.

| M15291/11 dash number | Type designation | For replacement only In new design, use M15291/9 | Circuit configuration |
|---|---|---|---|
| -001 | 20SR2A1 | M15291/12-001 | A (off-on-off-on) |
| -002 | 20SR3A1 | M15291/12-002 | A (off-on-off-on) |
| -003 | 20SR2B1 | M15291/12-003 | B (off-on 1-off-on 2) |
| -004 | 20SR3B1 | M15291/12-004 | B (off-on 1-off-on 2) |
| -005 | 20SR2E1 | M15291/12-005 | E (off-on 1-on 2-on 3) |
| -006 | 20SR3E1 | M15291/12-016 | E (off-on 1-on 2-on 3) |
| -007 | 20SR4F1 | M15291/12-007 | F (off-on 1-on 1 & 2-on 2) |
| -008 | 20SR6F1 | M15291/12-008 | F (off-on 1-on 1 & 2-on 2) |

Referenced Documents:

MIL-DTL-15291
MIL-DTL-15291/12
MIL-DTL-901
MIL-STD-167-1

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Navy - SH
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Preparing activity:
DLA - CC

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